

Exercises 1-1 Answer

6.

$$(1) y = \sqrt{2 + x - x^2}.$$

$$D = \{x \in \mathbb{R} \mid -1 \leq x \leq 2\},$$

$$R_f = \left\{y \in \mathbb{R} \mid 0 \leq y \leq \frac{3}{2}\right\}.$$

$$(2) y = \arccos\left(\frac{2x}{1+x^2}\right).$$

$$D = \{x \mid x \in \mathbb{R}\},$$

$$R_f = \{y \in \mathbb{R} \mid 0 \leq y \leq \pi\}.$$

$$(3) y = \sqrt{x - x^2}.$$

$$D = \{x \in \mathbb{R} \mid 0 \leq x \leq 1\},$$

$$R_f = \left\{y \in \mathbb{R} \mid 0 \leq y \leq \frac{1}{2}\right\}.$$

$$7. f\left(x + \frac{1}{x}\right) = x^2 + \frac{1}{x^2}.$$

$$t = x + \frac{1}{x} \in (-\infty, -2] \cup [2, +\infty),$$

$$t^2 = x^2 + 2 + \frac{1}{x^2},$$

$$f(t) = t^2 - 2 \Rightarrow f(x) = x^2 - 2, x \in (-\infty, -2] \cup [2, +\infty).$$

$$8. f\left(\frac{1}{x}\right) = x^2 + \sqrt{1 + x^2} (x > 0).$$

$$\begin{aligned} f(x) &= f\left(\frac{1}{\frac{1}{x}}\right) = \frac{1}{x} + \sqrt{1 + \frac{1}{x^2}} \\ &= \frac{1 + \sqrt{x^2 + 1}}{x}. \\ \Rightarrow f(x) &= \frac{1 + \sqrt{x^2 + 1}}{x}, x > 0. \end{aligned}$$

19.

(3)

$$f^{-1}(x) = \begin{cases} x, & x < 1 \\ \sqrt{x}, & 1 \leq x \leq 4 \\ \log_2 x, & x > 4 \end{cases}$$

(4)

$$y = \sqrt[3]{x + \sqrt{1+x^2}} + \sqrt[3]{x - \sqrt{1+x^2}},$$

$$y^3 = x + \sqrt{1+x^2} + x - \sqrt{1+x^2} + 3\sqrt[3]{(x + \sqrt{1+x^2})(x - \sqrt{1+x^2})} \left(\sqrt[3]{x + \sqrt{1+x^2}} + \sqrt[3]{x - \sqrt{1+x^2}} \right),$$

$$\Rightarrow y^3 = 2x - 3y,$$

$$\Rightarrow 3y^2 \frac{dy}{dx} + 3 \frac{dy}{dx} - 2 = 0,$$

$$\Rightarrow \frac{dy}{dx} = \frac{2}{3+3y^2} > 0.$$

$$x^3 = 2y - 3x,$$

$$y = \frac{1}{2}x^3 + \frac{3}{2}x.$$

$$f^{-1}(x) = \frac{1}{2}x^3 + \frac{3}{2}x, x \in \mathbb{R}$$

20.

$$\varphi(\varphi(x)) = \begin{cases} 0, & x \leq 0 \\ x, & x > 0 \end{cases},$$

$$\psi(\psi(x)) = 0,$$

$$\varphi(\psi(x)) = 0,$$

$$\psi(\varphi(x)) = \begin{cases} 0, & x \leq 0 \\ -x^2, & x > 0 \end{cases}.$$

23.

(2)even.

$$x \in \mathbb{R}, f(x) = \sqrt[3]{(1-x)^2} + \sqrt[3]{(1+x)^2},$$

$$x \in \mathbb{R}, f(-x) = \sqrt[3]{(1+x)^2} + \sqrt[3]{(1-x)^2} = f(x).$$

(4)odd.

$$f(x) = \ln \frac{1-x}{1+x}, \frac{1-x}{1+x} > 0 \Rightarrow x \in (-1, 1),$$

$$f(-x) = \ln \frac{1+x}{1-x}, \frac{1+x}{1-x} > 0 \Rightarrow x \in (-1, 1),$$

$$f(-x) + f(x) = \ln \frac{1+x}{1-x} \cdot \frac{1-x}{1+x} = \ln 1 = 0.$$

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习题 1-1 草稿

6 (1) $y = \sqrt{2+x-x^2}$

Domain $2+x-x^2 \geq 0$

$x \in [-1, 2]$

Range $0 \leq 2+x-x^2 \leq \sqrt{2+\frac{1}{2}-\frac{1}{4}} = \frac{3}{2}$

Range $y = [0, \frac{3}{2}]$

(2) $y = \arccos \frac{2x}{1+x^2}$

Domain $-1 \leq \frac{2x}{1+x^2} \leq 1$

$\begin{cases} -x^2 - 1 \leq 2x \\ 2x \leq x^2 + 1 \end{cases} \Rightarrow x \in \mathbb{R}$

Range $\frac{2x}{1+x^2} \stackrel{\text{t}}{=} \frac{2}{\frac{1}{x}+x} \in [-1, 1]$

Range $y = [0, \pi]$

(3) $y = \sqrt{x-x^2}$

Domain $x-x^2 \geq 0 \quad x \in [0, 1]$

Range $0 \leq x-x^2 \leq \frac{1}{2} - \frac{1}{4} = \frac{1}{4}$

Range $y = [0, \frac{1}{2}]$

7 $f(x + \frac{1}{x}) = x^2 + \frac{1}{x^2}$

$x + \frac{1}{x} \in (-\infty, -2] \cup [2, +\infty)$

let $x + \frac{1}{x} = t$

$t^2 = x^2 + 2 + \frac{1}{x^2} = x^2 + \frac{1}{x^2} + 2$

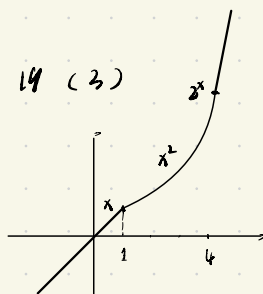
$\Rightarrow f(t) = t^2 - 2$

$f(x) = x^2 - 2, \quad x \in (-\infty, -2] \cup [2, +\infty)$

8 $f(\frac{1}{x}) = x + \sqrt{1+x^2} \quad (x > 0)$

$$f\left(\frac{1}{x}\right) = f(x) = \frac{1}{x} + \sqrt{1 + \frac{1}{x^2}}$$
$$= \frac{1 + \sqrt{1+x^2}}{x}$$

$\Rightarrow f(x) = \frac{1 + \sqrt{1+x^2}}{x}, \quad x > 0$



14 (3)

i) $x < 1$

$f^{-1}(x) = x$

ii) $1 \leq x \leq 4$

$x = y^2$

$\Rightarrow y = \sqrt{x}$

$f^{-1}(x) = \sqrt{x}$

iii) $x > 4$

$x = 2^y$

$\Rightarrow y = \log_2 x$

$f^{-1}(x) = \log_2 x$

$$f^{-1}(x) = \begin{cases} x, & x < 1 \\ \sqrt{x}, & 1 \leq x \leq 4 \\ \log_2 x, & x > 4 \end{cases}$$

$y^3 = 2x - 3 \left[(x + \sqrt{1+x^2})^{\frac{1}{3}} + (x - \sqrt{1+x^2})^{\frac{1}{3}} \right]$

(4) $y = \sqrt[3]{x + \sqrt{1+x^2}} + \sqrt[3]{x - \sqrt{1+x^2}}$

$x = \sqrt[3]{y + \sqrt{1+y^2}} + \sqrt[3]{y - \sqrt{1+y^2}}$

$$x^3 = y + \sqrt{1+y^2} + y - \sqrt{1+y^2} + 3[(y + \sqrt{1+y^2})(y - \sqrt{1+y^2})]^{\frac{1}{3}}$$
$$+ [(y + \sqrt{1+y^2})^{\frac{1}{3}} + (y - \sqrt{1+y^2})^{\frac{1}{3}}]^3$$

$y^3 = 2x - 3y$

$y^3 + 3y - 2x = 0$

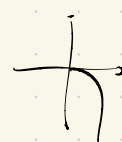
$2y^2 \frac{dy}{dx} + 3 \frac{dy}{dx} = 2 \Rightarrow \frac{dy}{dx} = \frac{2}{3+2y^2}$

$x^3 = 2y - 3x$

$\Rightarrow y = \frac{1}{2}x^3 + \frac{3}{2}x$

$\frac{dy}{dx} = \frac{2}{3+2y^2} \Rightarrow f^{-1}(x) = \frac{1}{2}x^3 + \frac{3}{2}x$

$$20 \quad \varphi(x) = \begin{cases} 0 & x \leq 0 \\ x & x > 0 \end{cases}$$

$$\psi(x) = \begin{cases} 0 & x \leq 0 \\ -x^2 & x > 0 \end{cases}$$


$$\varphi(\varphi(x)) = \begin{cases} 0 & x \leq 0 \\ x & x > 0 \end{cases}$$

$$\psi(\psi(x)) = 0$$

$$\varphi(\psi(x)) = 0$$

$$\psi(\varphi(x)) = \begin{cases} 0 & x \leq 0 \\ -x^2 & x > 0 \end{cases}$$

$$23 \quad (2) \quad f(x) = \sqrt[3]{(1-x)^2} + \sqrt[3]{(1+x)^2}$$

$$f(-x) = \sqrt[3]{(1+x)^2} + \sqrt[3]{(1-x)^2} = f(x)$$

even

$$(4) \quad f(x) = \ln \frac{1-x}{1+x} \quad \frac{1-x}{1+x} > 0 \Rightarrow (-1, 1)$$

$$f(-x) = \ln \frac{1+x}{1-x} \quad \frac{1+x}{1-x} > 0 \Rightarrow (-1, 1)$$

$$f(-x) + f(x) = \ln \frac{1+x}{1-x} \cdot \frac{1-x}{1+x} = \ln 1 = 0$$

odd